

Rilassamento continuo

Algoritmi di ottimizzazione [266SM]

Lorenzo Castelli

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**UNIVERSITÀ
DEGLI STUDI
DI TRIESTE**

Given an *IP*

$$z = \max\{c(x) : x \in X \subseteq \mathbb{Z}^n\}$$

how can we prove that a given point x^* is optimal?

We need to address this question because we saw that solution enumeration may not be possible, whereas disregarding variables' integrality constraints may not provide useful information.

Therefore we need to find alternatives (i.e., algorithms) to solve an *IP*.

The most common approach to solve *IP* problems is to find sequences of bounds until they are “close enough”.

Upper bound

If z is the optimal value of an *IP* problem, an **upper bound** is a value $\bar{z}$ such that $\bar{z} \geq z$.

Lower bound

If z is the optimal value of an *IP* problem, a **lower bound** is a value $\underline{z}$ such that $\underline{z} \leq z$.

Ideally, we would like to find $\bar{z}$ and $\underline{z}$ such that $\underline{z} = z = \bar{z}$.

From a practical point of view, any algorithm will look for a decreasing sequence of upper bounds

$$\overline{z_1} > \overline{z_2} > \dots > \overline{z_s} \geq z$$

and an increasing sequence of lower bounds

$$\underline{z_1} < \underline{z_2} < \dots < \underline{z_t} \leq z$$

and stops when

$$\overline{z_s} - \underline{z_t} \leq \epsilon$$

where ϵ is an appropriate non-negative value.

Lower bound

Every feasible solution $\hat{x} \in X$ provides a lower (or primal) bound $\underline{z} = c(\hat{x}) \leq z$.

For the problem

$$\begin{aligned}\max Z &= 1.00x_1 + 0.64x_2 \\ 50x_1 + 31x_2 &\leq 250 \\ 3x_1 - 2x_2 &\geq -4 \\ x_1, x_2 &\geq 0 \text{ and integer.}\end{aligned}$$

we saw, by rounding the optimal linear solution, that $\hat{x} = (2, 4)$ is a feasible solution such $\underline{z} = c(\hat{x}) = 4.56 \leq z$.

Findings upper bounds could be less obvious.

The most common idea is to replace a “difficult” *IP* problem by a simpler optimisation problem, whose optimal value is at least as large as z .

The simpler problem can be obtained by “relaxation”, i.e., by

- enlarging the set of feasible solutions so that one optimises over a larger set,
- replacing the $\max$ objective function by a function that has the same or a larger value everywhere.

Definition

A problem $(RP) z^R = \max\{f(x) : x \in T \subseteq \mathbb{R}^n\}$ is a **relaxation** of $(IP) z = \max\{c(x) : x \in X \subseteq \mathbb{Z}^n\}$ if:

- (i) $X \subseteq T$, and
- (ii) $f(x) \geq c(x)$ for all $x \in X$.

Proposition. If RP is a relaxation of IP , $z^R \geq z$

If x^* is an optimal solution of IP , $x^* \in X \subseteq T$ and $z = c(x^*) \leq f(x^*)$.
As $x^* \in T$, $f(x^*)$ is a lower bound on z^R , and so $z \leq f(x^*) \leq z^R$. ■

Hence, z^R is an upper bound!

Definition

For the integer program $\max\{cx : x \in X = P \cap \mathbb{Z}^n\}$ with formulation $P = \{x \in \mathbb{R}_+^n : Ax \leq b\}$, the **linear programming relaxation** is the linear program $z^{LP} = \max\{cx : x \in P\}$.

Since $X = P \cap \mathbb{Z}^n \subseteq P$ and the objective function is unchanged, this is clearly a relaxation.

Original *IP* problem

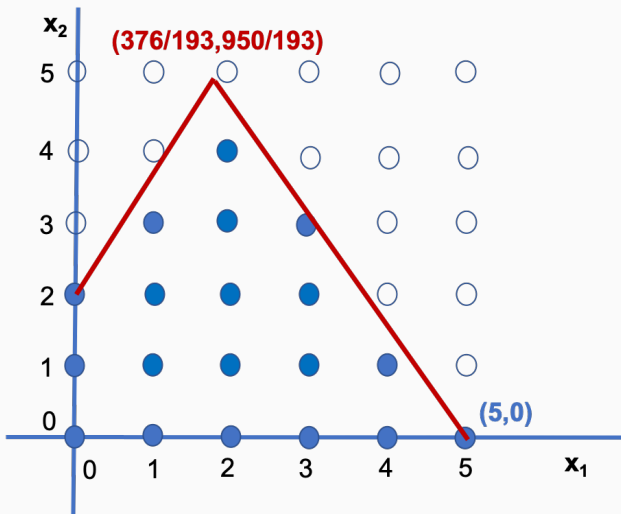
$$\begin{aligned} z &= \max 1.00x_1 + 0.64x_2 \\ 50x_1 + 31x_2 &\leq 250 \\ 3x_1 - 2x_2 &\geq -4 \\ x_1, x_2 &\geq 0 \text{ and integer.} \end{aligned}$$

LP relaxation

$$\begin{aligned} z^{LP} &= \max 1.00x_1 + 0.64x_2 \\ 50x_1 + 31x_2 &\leq 250 \\ 3x_1 - 2x_2 &\geq -4 \\ x_1, x_2 &\geq 0. \end{aligned}$$

- The optimal integer solution is $(5, 0)$ and $z = 5$
- The optimal solution of the linear relaxation is $(376/193, 950/193)$ and $z^{LP} = 984/193 = 5.098$.

Linear relaxation - example 1



For the *IP* problem

$$\begin{aligned} z &= \max 1.00x_1 + 0.64x_2 \\ 50x_1 + 31x_2 &\leq 250 \\ 3x_1 - 2x_2 &\geq -4 \\ x_1, x_2 &\geq 0 \text{ and integer.} \end{aligned}$$

We know that a lower bound is $\underline{z} = 4.560$ and an upper bound is $\bar{z} = 5.098$. The optimal value z therefore lies within

$$\underline{z} = 4.560 \leq z \leq 5.098 = \bar{z}.$$

In fact, $z = 5$. *The information we get by disregarding variables' integrality constraints can be very useful indeed.*

Consider the *IP* problem

$$z = \max 4x_1 - x_2$$

$$7x_1 - 2x_2 \leq 14$$

$$x_2 \leq 3$$

$$2x_1 - 2x_2 \leq 3$$

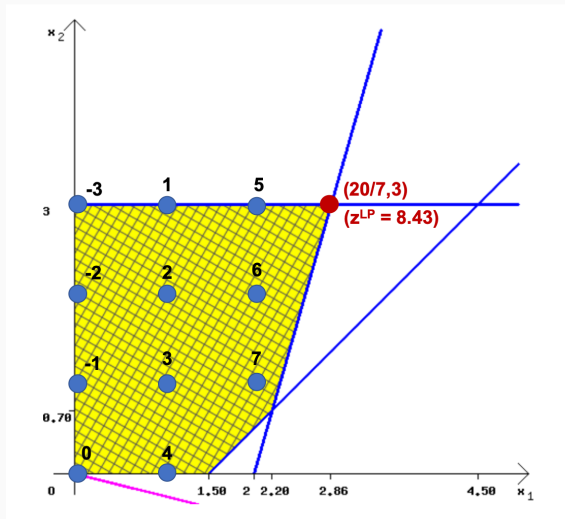
$$x_1, x_2 \geq 0 \text{ and integer.}$$

It is easy to see that $(2, 1)$ is a feasible solution, thus leading to the lower bound $\underline{z} = 7$. The optimal solution of the linear relaxation is $x^* = (20/7, 3)$ producing the upper bound $\bar{z} = 59/7 = 8.43$. Since **all coefficients of the objective function are integer** (i.e., $(4, -1)$) the optimal value z must to be integer as well. Hence, we can take as upper bound $\bar{z} = \lfloor 8.43 \rfloor = 8$. Therefore

$$\underline{z} = 7 \leq z \leq 8 = \bar{z}$$

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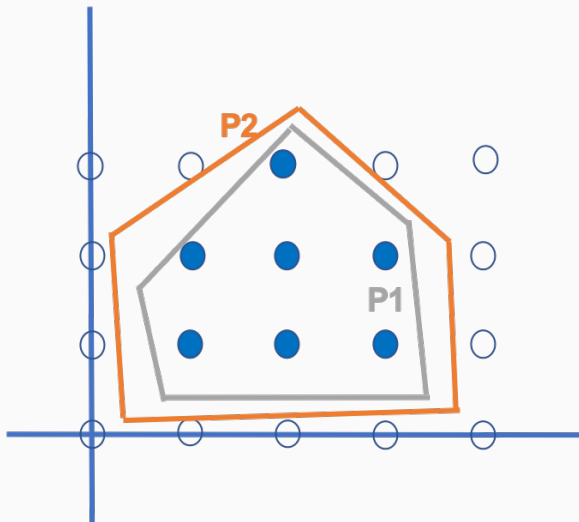
Linear relaxation - example 2



Proposition

Suppose P_1, P_2 are two formulations for the IP
 $\max\{cx : x \in X \subseteq \mathbb{Z}^n\}$ with P_1 a better formulation than P_2 , i.e.,
 $P_1 \subset P_2$. If $z_i^{LP} = \max\{cx : x \in P_i\}$ for $i = 1, 2$ are the values of the
associated linear programming relaxations, then $z_1^{LP} \leq z_2^{LP}$ for all c .

The better the formulation the tighter the upper bound.



If a relaxation RP is infeasible, the original problem IP is infeasible

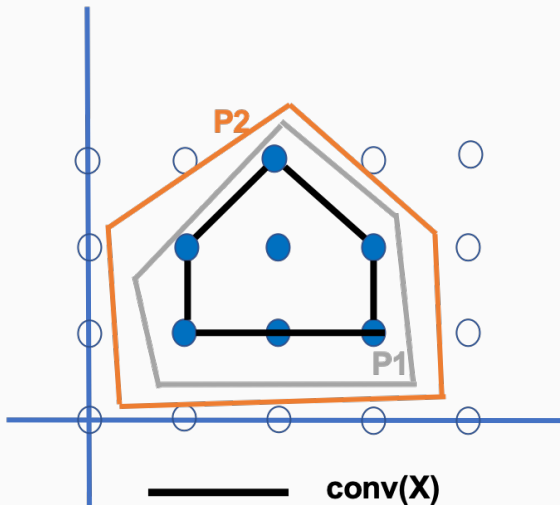
As RP is infeasible, $T = \emptyset$. Since $X \subseteq T$ also $X = \emptyset$.

Let x^* be an optimal solution of RP . If $x^* \in X$ and $f(x^*) = c(x^*)$ then x^* is an optimal solution of IP .

As $x^* \in X$, $z \geq c(x^*) = f(x^*) = z^R$. As $z \leq z^R$, we obtain $c(x^*) = z = z^R$.

- If the optimal solution x^* of the linear relaxation is integer, x^* is also the optimal solution of the original IP problem.
- The optimal solution of $conv(x)$ is always integer because all extreme points belong to it.

Convex hull - integer solution



Example 1

$$\begin{aligned} \max & 7x_1 + 4x_2 + 5x_3 + 2x_4 \\ & 3x_1 + 3x_2 + 4x_3 + 2x_4 \leq 6 \\ & x_i \in \{0, 1\} \text{ for } i = 1, 2, 3, 4. \end{aligned}$$

The linear relaxation has optimal solution $x^* = (1, 1, 0, 0)$. Hence, x^* is the optimal solution of the original binary problem.

Example 2

$$\begin{aligned} \max & 3x_1 + 5x_2 \\ & x_1 \leq 4 \\ & 2x_2 \leq 12 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0 \text{ and integer} \end{aligned}$$

The linear relaxation has optimal solution $x^* = (2, 6)$. Hence, x^* is the optimal solution of the original integer problem.

In case of the minimisation problem IP

$$z = \min\{cx : x \in X \subseteq \mathbb{Z}^n\}$$

a feasible solution of X identifies an upper bound for z , while a relaxation of IP identifies a lower bound for z . In this case, a relaxation exists if $f(x) \leq c(x)$ for all $x \in X$ - see slide (6).

Thank you for your attention

